CRIMEBUSTERS ANSWER KEY 2026
RALEIGH DIV. B INVITATIONAL
ALL ANSWERS IN BOLD & ITALICS

Section B - Identification of Substances

Powders			
Identification	*Name* (2 Points each)	*Formula* (2 Points each)	*Who does this point to* (1 Point each)
S1	***Baking Soda***	$NaHCO_3$	***Mrs. Monoxide and/or Dasan***
S2	***Table Salt***	$NaCl$	***Mrs. Monoxide and/or Dasan***
S3	***Flour***	$C_6H_{10}O_5$	***Mrs. Monoxide***
S4	***Sugar***	$C_{12}H_{22}O_{11}$	***Mrs. Monoxide***
S5	***Alka-Seltzer***	$C_8H_{12}KNaO_{13}^{2+}$	***Dasan***
S6	***Sand***	SiO_2	***Agua de Fina***
S7	***Corn Starch***	$C_6H_{10}O_5$	***Mrs. Monoxide and/or Dasan***
S8	***Yeast***	***N/A***	***Mrs. Monoxide***
S9	***Powdered Limestone or Calcium Carbonate***	$CaCO_3$	***None***
Liquids			
Identification	*Name* (2 Points each)	*Formula* (2 Points each)	*Who does this point to* (1 Point each)
L1	***Lemon Juice***	$C_6H_8O_7$	***Mrs. Monoxide***

L2	***Vinegar***	CH_3COOH	***Dasan***
L3	***Water***	H_2O	***Agua de Fina***

Section C - Polymer Testing

Plastics				
Identification	*Description/Test Results* (1 Point each)	*Plastic Name* (2 Point each)	*Formula* (2 Points each)	*Who does this point to?* (1 Point each)
P1	***Sinks with all tests***	***PETE***	$C_{10}H_8O_4$	***Used to make food containers and plastic packaging, Mrs. Monoxide and Agua de Fina***
P2	***Floats with Water and all concentrations of NaCl*** ***Sinks with Alcohol and corn oil***	***HDPE***	C_2H_4	***Household items and packaging, Mrs. Monoxide***
Hair				
Identification	*Picture*	*Hair Name* (3 Points each)	*Who does this point to?* (1 Point each)	
H1		**Cat Hair**	***Mrs. Monoxide***	
H2		***Human Hair***	***None***	
Fibers				
Identification	*Cloth Microscope Image*	*Fiber Name* (3 Points)	*Who does this point to?* (1 Point)	
C1		***Animal/Cotton***	***Many towels and cloth cooking supplies are cotton, so it could lead to Mrs. Monoxide, but it could also point to no one***	

Section D - Chromatography
Both Dasan and Mrs. Monoxide have a “signature” pen they like to use. Dasan uses it to take notes on finances, while Mrs. Monoxide focuses more on inks that can survive in the kitchen. A chromatography test was done on both pens.

Whose pen is it? (5 Points)
Mrs. Monoxide

Section E - Physical Evidence
DNA was found on the victims neck. Find out whose DNA it was.

Whose DNA is this? (5 Points)

Mrs. Monoxide

Section F - Analysis

1. Who committed the crime? (10 Points)
   Mrs. Monoxide

2. Analyze your evidence pointing to them and why they did it. (30 Points)
   a. ***Most evidence points to her (10 Points)***
   b. ***Describing the evidence pointing to her (10 Points)***
   c. ***List motives (10 Points)***